Itamar Pres

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Education

Massachusetts Institute of Technology, EECS — Cambridge, MA
PhD in Computer Science (Advised by **Jacob Andreas**)

UNIVERSITY OF MICHIGAN ANN ARBOR, College of Engineering — Ann Arbor, MI
BSE in Computer Science with minor in Mathematics: Summa Cum Laude (GPA: 3.83) — *May 2025*

Research Publications

Self-CTRL: Self-Consistency Training with Reinforcement Learning *2026*
Pres, Ruis, Ghebreselassie, Li, Andreas
Preprint at Actionable Interpretability Workshop COLM 2026 ***(Oral: top 4 submissions)***

- Introduced a method that optimizes for consistency between a model's self-explanations of behavior and its behavior
- Improved a third-party auditor's refusal-prediction accuracy from 36% to 92%

Position: It's Time to Optimize for Self-Consistency *2026*
Pres*, Li*, Ruis*, et al.
The Forty-third International Conference on Machine Learning

- Proposed self-consistency as a framework for understanding diverse LM failures, including sycophancy
- Showed that many existing techniques are special cases of a common consistency-optimization procedure

Position: Why LLMs Should Be Reasonably Morally Inconsistent *2026*
Stenseke, Kierans, **Pres**, Hadfield-Menell
The Forty-third International Conference on Machine Learning - Workshop on Pluralistic Alignment

- Argued that **moral consistency** should be a desirable but defeasible norm, with deviations made transparent

Competition Dynamics Shape Algorithmic Phases of In-Context Learning *2025*
Park*, Lubana*, **Pres**, et al.
The 13th International Conference on Learning Representations ***(Spotlight)***

- Provided mechanistic analysis showing Transformers solve in-context learning with multiple competing algorithms
- Developed quantifiable techniques to measure neuron memorization of training data in transformers

Towards Reliable Evaluation of Behavior Steering Interventions in LLMs *2024*
Pres, et al.
The 38th Conference on Neural Information Processing Systems - Workshop on Foundation Model Interventions ***(Oral: top 5 submission)***

- Established four criteria for effective behavioral steering metrics and highlighted current metric shortcomings
- Developed new steering metric showing over-reported success of control method Contrastive Activation Steering

A Mechanistic Understanding of Alignment Algorithms: A Case Study on DPO and Toxicity *2024*
Lee, Bai, **Pres**, et al.
The Forty-first International Conference on Machine Learning ***(Oral: top 1.5% of submissions)***

- Discovered how toxicity is represented in pre-trained language models (GPT2, Llama2-7b)
- Studied the internal mechanisms of a popular alignment algorithm (Direct Preference Optimization)
- Discovered that DPO does not remove toxicity representations but rather suppresses them
- Uncovered why "aligned" models can be jailbroken

Previous Research Experience

Krueger AI Safety Lab Internship (University of Cambridge) — *May 2024 - Dec 2024*
Mentored By ***David Krueger***

- Developed framework for multiple constraint satisfaction of Large Language Model generations at inference-time
- Explored generalizability of inference-time interventions such as Contrastive-Activation-Steering
- Developed novel benchmark to measure the success of behavioral steering interventions

Physics of Intelligence at the Center for Brain Science (Harvard University) — *Sept 2024 - May 2025*
Mentored By ***Hidenori Tanaka, Ekdeep Singh Lubana***

- Rigorously studying Transformer model behaviors in thoughtfully constructed toy-settings
- Trained Transformers of different architectures to perform in-context learning (ICL)
- Researching the mechanisms these Transformers use to solve ICL tasks and handle underspecified queries
- Published findings at ICLR

University of Michigan Language and Information Technologies Lab *October 2023 - May 2025*

Supervised by ***Rada Mihalcea***

- Conducted research in mechanistic interpretability, with a focus on safety alignment algorithms
- Published research findings at ICML
- Researched Vision Language Model interpretability as well as In Context Learning mechanisms

Mechanistic Interpretability Research (MATS) *June 2023 – Aug 2023*

Mentored by ***Neel Nanda***

- Analyzed and visualized high-dimensional LLM operations to generate hypothesis of LLM internal mechanisms
- Tested hypotheses of attention head superposition phenomenon in GPT2-XL
- Devised precise casual interventions to validate hypotheses under fact extraction contexts

Grants and Awards

National Science Foundation Graduate Research Fellowship (NSF GRFP) *April 2026*

2024-25 University of Michigan Outstanding Undergraduate Research Award *May 2025*

Effective Altruism Infrastructure Grant, Full stack Development *June 2022 – September 2022*